

Solve the following equation:

$$\log_2(x) + \log_2(x-4) = 3$$

$$\log_2(x(x-4)) = 3$$

$$\frac{\ln(x(x-4))}{\ln 2} = 3$$

$$\ln(x(x-4)) = 3 \ln 2$$

$$x^2 - 4x = 8$$

$$x^2 - 4x - 8 = 0$$

$$x_1 = 5.4641 \quad \text{and} \quad x_2 = -1.4641$$

accepted

rejected

$$\text{D.F. : } x > 4$$

$$\boxed{x = 5.4641}$$

